

Applications, Advantages and Disadvantages of Graph

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Graph is a non-linear data structure that contains nodes (vertices) and edges. A graph is a collection of set of vertices and edges (formed by connecting two vertices). A graph is defined as $G = \{V, E\}$ where V is the set of vertices and E is the set of edges.

Graphs can be used to model a wide variety of real-world problems, including social networks, transportation networks, and communication networks. They can be represented in various ways, such as by a set of vertices and a set of edges, or by a matrix or an adjacency list. The two most common types of graphs are directed and undirected graphs.

Terminologies of Graphs:

.An edge is one of the two primary units used to form graphs. Each edge has two ends, which are vertices to which it is attached.

.If two vertices are endpoints of the same edge, they are adjacent.

.A vertex's outgoing edges are directed edges that point to the origin.

.A vertex's incoming edges are directed edges that point to the vertex's destination.

.The total number of edges occurring to a vertex in a graph is its degree.

.A vertex with an in-degree of zero is referred to as a source vertex, while one with an out-degree of zero is known as sink vertex.

.A path is a set of alternating vertices and edges, with each vertex connected by an edge.

.The path that starts and finishes at the same vertex is known as a cycle.

.A path with unique vertices is called a simple path.

.A spanning subgraph that is also a tree is known as a spanning tree.

.A connected component is the unconnected graph's most connected subgraph.

.A bridge, which is an edge of removal, would sever the graph.

.Forest is a graph without a cycle.

Graph Representation:

Graph can be represented in the following ways:

1. **Set Representation:** Set representation of a graph involves two sets: Set of vertices $V = \{V_1, V_2, V_3, V_4\}$ and set of edges $E = \{\{V_1, V_2\}, \{V_2, V_3\}, \{V_3, V_4\}, \{V_4, V_1\}\}$. This representation is efficient for memory but does not allow parallel edges.
2. **Sequential Representation:** This representation of a graph can be represented by means of matrices: Adjacency Matrix, Incidence matrix and Path matrix.
 - **Adjacency Matrix:** This matrix includes information about the adjacent nodes. Here, $a_{ij} = 1$ if there is an edge from V_i to V_j otherwise 0 . It is a matrix of order $V \times V$.
 - **Incidence Matrix:** This matrix includes information about the incidence of edges on the nodes. Here, $a_{ij} = 1$ if the j^{th} edge E_j is incident on i^{th} vertex V_i otherwise 0 . It is a matrix of order $V \times E$.
 - **Path Matrix:** This matrix includes information about the simple path between two vertices. Here, $P_{ij} = 1$ if there is a path from V_i to V_j otherwise 0 . It is also called as reachability matrix of graph G .
3. **Linked Representation:** This representation gives the information about the nodes to which a specific node is connected i.e. adjacency lists. This representation gives the adjacency lists of the vertices with the help of array and linked lists. In the adjacency lists, the vertices which are connected with the specific vertex are arranged in the form of lists which is connected to that vertex.

Real-Time Applications of Graph:

- **Social media analysis:** Social media platforms generate vast amounts of data in real-time, which can be analyzed using graphs to identify trends, sentiment, and key influencers. This can be useful for marketing, customer service, and reputation management.

- **Network monitoring:** Graphs can be used to monitor network traffic in real-time, allowing network administrators to identify potential bottlenecks, security threats, and other issues. This is critical for ensuring the smooth operation of complex networks.
- **Financial trading:** Graphs can be used to analyze real-time financial data, such as stock prices and market trends, to identify patterns and make trading decisions. This is particularly important for high-frequency trading, where even small delays can have a significant impact on profits.
- **Internet of Things (IoT) management:** IoT devices generate vast amounts of data in real-time, which can be analyzed using graphs to identify patterns, optimize performance, and detect anomalies. This is important for managing large-scale IoT deployments.
- **Autonomous vehicles:** Graphs can be used to model the real-time environment around autonomous vehicles, allowing them to navigate safely and efficiently. This requires real-time data from sensors and other sources, which can be processed using graph algorithms.
- **Disease surveillance:** Graphs can be used to model the spread of infectious diseases in real-time, allowing health officials to identify outbreaks and implement effective containment strategies. This is particularly important during pandemics or other public health emergencies.
- The best example of graphs in the real world is Facebook. Each person on Facebook is a node and is connected through edges. Thus, A is a friend of B. B is a friend of C, and so on.

Advantages of Graph:

- **Representing complex data:** Graphs are effective tools for representing complex data, especially when the relationships between the data points are not straightforward. They can help to uncover patterns, trends, and insights that may be difficult to see using other methods.
- **Efficient data processing:** Graphs can be processed efficiently using graph algorithms, which are specifically designed to work with graph data structures. This makes it possible to perform complex operations on large datasets quickly and effectively.

- **Network analysis:** Graphs are commonly used in network analysis to study relationships between individuals or organizations, as well as to identify important nodes and edges in a network. This is useful in a variety of fields, including social sciences, business, and marketing.
- **Pathfinding:** Graphs can be used to find the shortest path between two points, which is a common problem in computer science, logistics, and transportation planning.
- **Visualization:** Graphs are highly visual, making it easy to communicate complex data and relationships in a clear and concise way. This makes them useful for presentations, reports, and data analysis.
- **Machine learning:** Graphs can be used in machine learning to model complex relationships between variables, such as in recommendation systems or fraud detection.
- Graphs are used in computer science to depict the flow of computation.
- Users on Facebook are referred to as vertices, and if they are friends, there is an edge connecting them. The Friend Suggestion system on Facebook is based on graph theory.
- You come across the Resources Allocation Graph in the Operating System, where each process and resource are regarded vertically. Edges are drawn from resources to assigned functions or from the requesting process to the desired resources. A stalemate will develop if this results in the establishment of a cycle.
- Web pages are referred to as vertices on the World Wide Web. Suppose there is a link from page A to page B that can represent an edge. this application is an illustration of a directed graph.
- Graph transformation systems manipulate graphs in memory using rules, Graph databases store and query graph-structured data in a transaction-safe, permanent manner.

Disadvantages of Graph:

- **Limited representation:** Graphs can only represent relationships between objects, and not their properties or attributes. This means that in order to fully understand the data, it may be necessary to supplement the graph with additional information.

- **Difficulty in interpretation:** Graphs can be difficult to interpret, especially if they are large or complex. This can make it challenging to extract meaningful insights from the data, and may require advanced analytical techniques or domain expertise.
- **Scalability issues:** As the number of nodes and edges in a graph increases, the processing time and memory required to analyze it also increases. This can make it difficult to work with large or complex graphs.
- **Data quality issues:** Graphs are only as good as the data they are based on, and if the data is incomplete, inconsistent, or inaccurate, the graph may not accurately reflect the relationships between objects.
- **Lack of standardization:** There are many different types of graphs, and each has its own strengths and weaknesses. This can make it difficult to compare graphs from different sources, or to choose the best type of graph for a given analysis.
- **Privacy concerns:** Graphs can reveal sensitive information about individuals or organizations, which can raise privacy concerns, especially in social network analysis or marketing.

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